Chapter 5
Differentiation

Note 2

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5.2: Differentiability Theorems

There are a number of basic properties of the derivative that are very useful in the calculation of the derivatives of various combinations of functions. Justification of some of these properties are provided, which will be familiar to the you from earlier courses.

Theorem 5.2.1 : Let f and g be real functions and $\alpha \in \mathbb{R}$. If f and g are differentiable at a, then $f+g,\ \alpha f,\ f\cdot g$, and [when $g(a)\neq 0$] f/g are all differentiable at a. In fact,

$$(f+g)'(a) = f'(a) + g'(a)\ , \tag{5}$$

$$(\alpha f)'(a) = \alpha f'(a)\ , \tag{6}$$

$$(f\cdot g)'(a) = g(a)f'(a) + f(a)g'(a)\ , \tag{7}$$

$$\left(\frac{f}{g}\right)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{g^2(a)}\ . \tag{8}$$

Proof of Theorem 5.2.1.

- We provide the details only for (7) and the rest are left as Exercise.
- By adding and subtracting $f(a)g(x)$ in the numerator of the left side of the following expression,we can write
$$\frac{f(x)g(x) - f(a)g(a)}{x-a} = g(x)\ \frac{f(x)-f(a)}{x-a} + f(a)\ \frac{g(x)-g(a)}{x-a}.$$
- Since g is continuous, and from the definition of derivative, follows from that
$$\lim_{x\to a} \frac{f(x)g(x) - f(a)g(a)}{x-a} = g(a)f'(a) + f(a)g'(a).$$

□

Note : Formula (5) is called the Sum Rule, (6) is sometimes called the Homogeneous Rule, (7) is called the Product Rule, and (8) is called the Quotient Rule.

Next, we show what the derivative does to a composition of two functions.

Theorem 5.2.2 : Chain Rule

Let f and g be real functions. If f is differentiable at a and g is differentiable at $f(a)$, then $g \circ f$ is differentiable at a with

$$(g \circ f)'(a) = g'(f(a))f'(a) \ . \tag{9}$$

Proof of Theorem 5.2.2.

- By **Theorem 5.1.1**, there exist open intervals I and J, and functions $F : I \to \mathbb{R}$, continuous at a, and $G : J \to \mathbb{R}$, continuous at $f(a)$, such that $F(a) = f'(a)$, $G(f(a)) = g'(f(a))$,

  $$f(x) = F(x)(x - a) + f(a), \quad x \in I \ , \tag{10}$$

  and

  $$g(y) = G(y)(y - f(a)) + g(f(a)), \quad y \in J \ . \tag{11}$$

- Since f is continuous at a, we may assume (by making I smaller if necessary) that $f(x) \in J$ for all $x \in I$.

- Fix $x \in I$. Apply (11) to $y = f(x)$ and (10) to x to write

  $$\begin{aligned}(g \circ f)(x) = g(f(x)) &= G(f(x))(f(x) - f(a)) + g(f(a)) \\ &= G(f(x))F(x)(x - a) + (g \circ f)(a).\end{aligned}$$

- Set $H(x) = G(f(x))F(x)$ for $x \in I$.

- Since F is continuous at a and G is continuous at $f(a)$, it is clear that H is continuous at a.

- Moreover, we have

  $$H(a) = G(f(a))F(a) = g'(f(a))f'(a).$$

- Therefore, it follows from **Theorem 5.1.1** that $(g \circ f)'(a) = g'(f(a))f'(a)$.

□

EXERCISES

(1). Suppose that I is an open interval containing a, and $f, g, h : I \to \mathbb{R}$. Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

(a) If f, g and h are differentiable at a, then

$$(fgh)'(a) = f'(a)g(a)h(a) + f(a)g'(a)h(a) + f(a)g(a)h'(a).$$

(b) If f is differentiable at a and g is differentiable at $f(a)$, then

$$(g \circ f)''(a) = g'(f(a))f''(a) + g''(f(a))(f'(a))^2.$$

(c) If the nth-order derivatives $f^{(n)}(a)$ and $g^{(n)}(a)$ exist, then

$$(f+g)^{(n)}(a) = f^{(n)}(a) + g^{(n)}(a).$$

(d) If the nth-order derivatives $f^{(n)}(a)$ and $g^{(n)}(a)$ exist and are nonzero, then

$$\left(\frac{f}{g}\right)^{(n)}(a) = \frac{g(a)f^{(n)}(a) + (-1)^n f(a)g^{(n)}(a)}{g^{n+1}(a)}.$$

(2). (**POWER RULE**) Assume that $(e^x)' = e^x$ for $x \in \mathbb{R}$ and $(\log x)' = \dfrac{1}{x}$ for $x > 0$. Use $x^\alpha := e^{\alpha \log x}$ to prove that $(x^\alpha)' = \alpha x^{\alpha-1}$ for all $x > 0$ and $\alpha \in \mathbb{R}$.

(3). Suppose that f and g are differentiable at 2 and 3 with $f'(2) = a,\ f'(3) = b,\ g'(2) = c$, and $g'(3) = d$. If $f(2) = 1,\ f(3) = 2,\ g(2) = 3$, and $g(3) = 4$, evaluate each of the following derivatives.

(a) $(fg)'(2)$ (b) $\left(\dfrac{f}{g}\right)'(3)$ (c) $(g \circ f)'(3)$ (d) $(f \circ g)'(2)$

(4). Suppose that f is differentiable at 2 and 4 with $f(2) = 2,\ f(4) = 3,\ f'(2) = \pi$, and $f'(4) = e$.

(a) If $g(x) = xf(x^2)$, find the value of $g'(2)$.

(b) If $g(x) = f^2(\sqrt{x})$, find the value of $g'(4)$.

(c) If $g(x) = \dfrac{x}{f(x^3)}$, find the value of $g'(\sqrt[3]{2})$.

(5). Suppose that f is differentiable at a and $f(a) \neq 0$.

(a) Show that for h sufficiently small, $f(a+h) \neq 0$.

(b) (**RECIPROCAL RULE**) Using **Definition 6.1.1** directly, prove that $\dfrac{1}{f(x)}$ is differentiable at $x = a$ and

$$\left(\frac{1}{f}\right)'(a) = -\frac{f'(a)}{f^2(a)}.$$

(c) Use the Product Rule and Reciprocal Rule to prove the Quotient Rule directly.

(6). Assuming that e^x is differentiable on $\mathbb{R}$, prove that

$$f(x) = \begin{cases} \dfrac{x}{1+e^{1/x}} & : \ x \neq 0 \\ 0 & : \ x = 0 \end{cases}$$

is differentiable on $[0, \infty)$. Is f differentiable at 0?

(7). Suppose that $n \in \mathbb{N}$ and f, g are real functions of a real variable whose nth derivatives $f^{(n)},\ g^{(n)}$, exist at a point a. Prove Leibniz's generalization of the Product Rule:

$$(fg)^{(n)}(a) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(a) g^{(n-k)}(a).$$